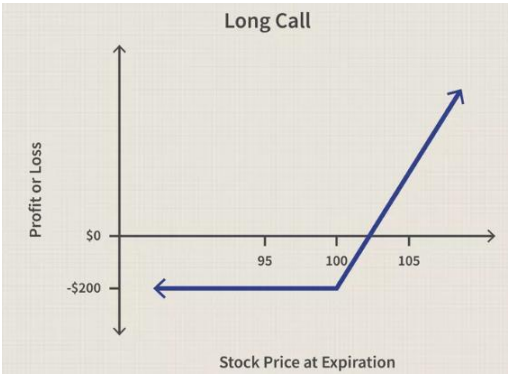
$$C = N(d_1)S_t - N(d_2)Ke^{-rt}$$



STAT 482 PROJECT

Modeling European Call Options

(Geometric Brownian Motion and Black-Scholes)

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Introduction

During the early pandemic in 2020 I briefly considered studying finance, including taking and narrowly failing one of the Chartered Financial Analyst Exams. I was keen to apply some of my limited knowledge of financial theory to at least one class at Long Beach, and Black-Scholes was an excellent chance to do so. Because other students elaborated on the Geometric Brownian Motion modeling process I focused more on the theory and meaning behind Black-Scholes itself in my presentation. We have decided to model the large S&P 500 company Netflix over the last 22 years of its stock price and predict the spread of its call option prices. The methodology here is that Netflix is a large, frequently traded company with many interested buyers and sellers, so the price is especially liquid and market efficient: a ripe candidate for a simple model. We are excited to learn how accurate the 1973 model is when applied to contemporary European Call Option pricing.

Background

1973 Fisher Black and Myron Scholes elaborated on Geometric Brownian motion with the Black-Scholes model, which revolutionized finance by providing a structured approach to derivative pricing, leading to the development of the options market. Black Scholes operates under assumptions including that the underlying asset price follows a lognormal distribution and evolves according to a geometric Brownian motion. It also assumes several financial axioms, including markets being efficient with no arbitrage opportunities, no transaction costs or taxes, and that the risk-free rate and volatility are constant. The base Black-Scholes model also works only on European Call Options which can only be exercised at expiration, unlike an American Option which can be exercised at any time and therefore requires a more complex model.

Data Description

We decided to model the Netflix stock and option market, using a dataset on Kaggle. The dataset contains columns for Date, Open Price, High Price, Low Price, Close Price, and Volume. The data set has an observation for every every market day starting with Netflix's IPO on May 23rd, 2002, to July 15th, 2024, for 5573 observations total. The price rose from 1.19 dollars per share in 2002 to 656.45 dollars per share in 2024.

Results

In R, we first plotted our historical stock price from 2002 to 2024, and plotted the distribution of log-price increments, which we were able to analyze visually and classify as normally distributed. We then proceeded to model our stock price with Geometric Brownian

Motion, which gave us a drift coefficient estimate μ of .001132 and a volatility estimate σ of .03554163.

Now that we know the estimated parameters of our stock distribution, we can use Black-Scholes formula to predict an individual European Call Option price. Where X(0) is our current stock price, t is the time until strike price, K is our option strike price, μ and σ are our estimated coefficients, and C is our estimated option price, the formula for our simplified estimated option price is

$$C = X(0)\,\Phi\big(A + \sigma\sqrt{t}\big) - e^{-rt}\,K\,\Phi(A)$$

where

$$A = \frac{\mu t - \ln(K/X(0))}{\sigma\sqrt{t}}$$

and

$$r = \mu + \frac{\sigma^2}{2}$$

If we are particularly bullish and are interested in buying a European Call Option with the Netflix stock as the underlying, we could predict the price of one such option and compare it to the actual market. If we want to predict the price of a European Call Option that vests in 30 days at a stock price of 950 dollars (the current stock price is 902.17 dollars) then we plug out 5 inputs into our cost function and solve:

$$A = \frac{\mu t - \ln(K/X(0))}{\sigma\sqrt{t}} = -.0909227$$

$$r = \mu + \frac{\sigma^2}{2} = .0017636$$

and therefore

$$C = X(0)\,\Phi\big(A + \sigma\sqrt{t}\big) - e^{-rt}\,K\,\Phi(A)$$

= 902.17 ϕ (.10373) – 901.0439 ϕ (-.0909227)

= 70.47 dollars for European call option expiring in 30 days at $ 950.

Conclusion

While our Black Scholes model predicted a price of 70.47 dollars for the call option, when we compare it to the real-world market we were surprisingly far off. See below picture of the yahoo finance board, which contains a list of Netflix call options at a strike price of 950. Note the call options don't reach a price of ~70 dollars until approximately the month of May (t=150) and the t=30 K=950 call options are only priced at 14-18.65 dollars.

Calls In The Money

Contract Name	Last Trade Date (EDT)	Expire Date	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
NFLX241206C00950000	12/5/2024 8:59 PM	2024-12-06	0.22	0.20	0.24	-0.18	-45.00%	954	1,296	36.13%
NFLX241213C00950000	12/5/2024 8:59 PM	2024-12-13	3.71	3.60	3.95	+0.41	+12.42%	4,803	803	26.80%
NFLX241220C00950000	12/5/2024 8:57 PM	2024-12-20	7.60	7.30	7.80	+0.85	+12.59%	384	2,767	26.31%
NFLX241227C00950000	12/5/2024 8:54 PM	2024-12-27	10.05	10.00	10.65	+0.95	+10.44%	92	237	25.45%
NFLX250103C00950000	12/5/2024 8:51 PM	2025-01-03	14.00	12.50	14.20	+1.65	+13.36%	42	49	26.01%
NFLX250110C00950000	12/5/2024 8:32 PM	2025-01-10	18.65	16.95	18.40	+3.35	+21.90%	18	27	27.29%
NFLX250117C00950000	12/5/2024 8:46 PM	2025-01-17	25.24	24.40	25.05	+2.99	+13.44%	419	1,033	30.53%
NFLX250221C00950000	12/5/2024 8:59 PM	2025-02-21	48.30	47.05	49.70	+3.45	+7.69%	75	365	37.49%
NFLX250321C00950000	12/5/2024 8:33 PM	2025-03-21	56.87	55.20	57.70	+3.17	+5.90%	50	324	36.23%

Prices on the Robinhood trading app were in a similar neighborhood to yahoo finance. I suspect that using the entire 22-year history of Netflix's stock price is less accurate than a more relevant time window because it overestimates structural volatility. Perhaps just using the last 10 years of data, where Netflix is a streaming, production, and technology company rather than the earlier days of the company, provides a better estimate of stock expectations. A second option for this disparity is that our model is just too simple, while a third option would be that there are current market forces affecting the short-term price of the option at a larger magnitude than our coefficients modeled with GBM. Nevertheless, we are pleased with our conclusions because they demonstrate how the fundamentals of Black Scoles still underline our modern options markets, and because they gave us the opportunity to apply concepts learned in our Stat 482 class.

Appendix

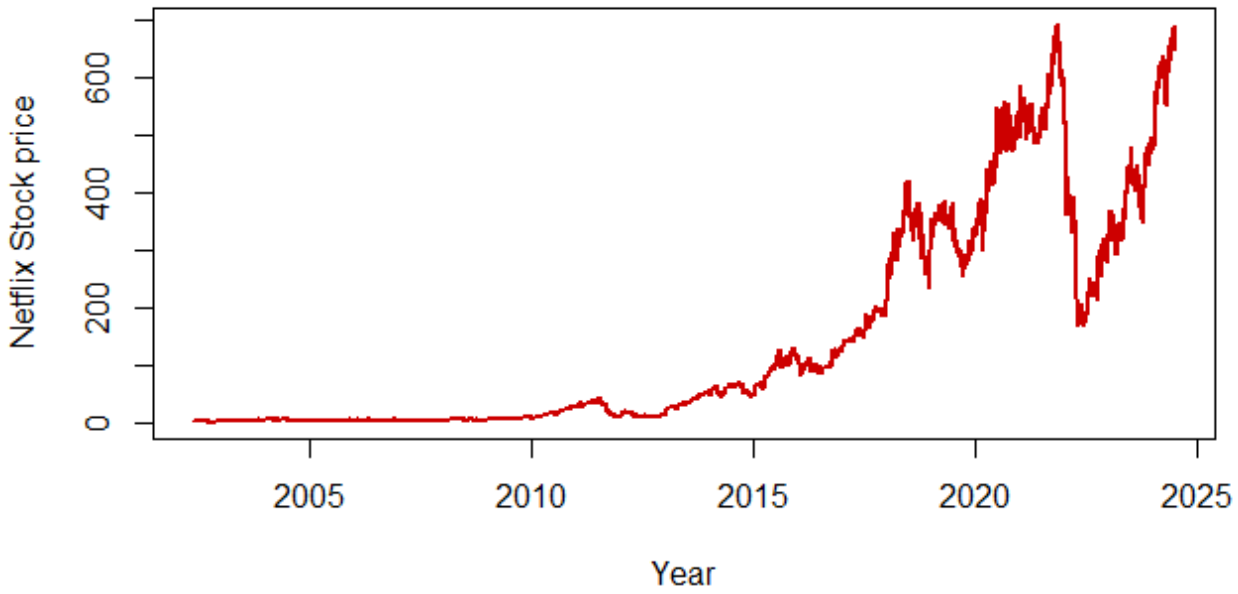
A. Geometric Brownian Motion

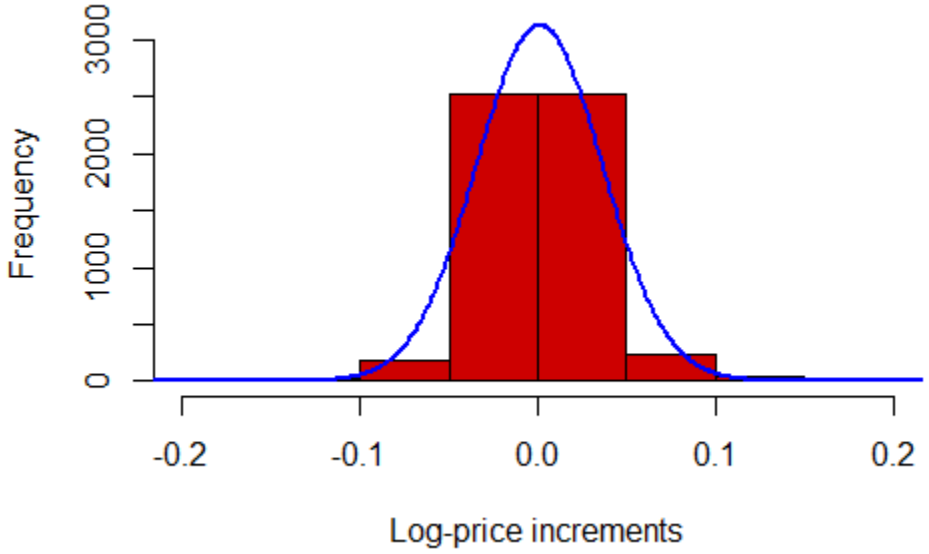
a) r studio code

```
stock.data<- read.csv(file="C:/Users/jknau/OneDrive/Desktop/482 Random Processes/NFL
str(stock.data$Date)
stock.data$Date <- as.Date(stock.data$Date, format="%m/%d/%Y")
str(stock.data$Date)
Date<- as.POSIXct(stock.data$Date)

price<- stock.data$Close
#plotting stock price against date
plot(Date, price, type="l", lwd=2, cex=0, col="red3",
     xlab="Year", ylab="Netflix Stock price")

#calculating increments of log-price
log.inc<- c()
price1<- price[-1]
price1.lag<- head(price, -1)
log.ratio<- log(price1/price1.lag)
#plotting histogram
library(rcompanion)

plotNormalHistogram(log.ratio, xlab="Log-price increments",
                    col="red3", xlim=c(-.2, .2))
```

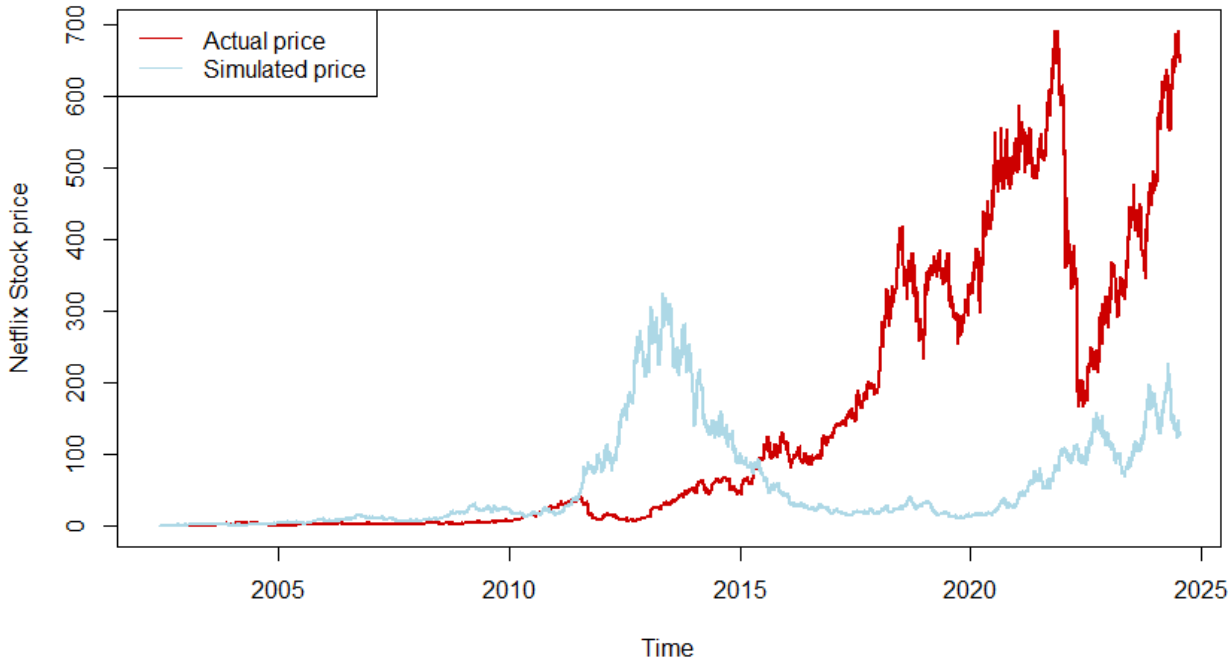
```
#estimating parameters
print(mu.hat<- mean(log.ratio))
print(sigma.hat<- sd(log.ratio))

#specifying Brownian motion as vector
BM<- c()
#specifying initial value
BM[1]<- 0
#specifying seed
set.seed(34235457)
#simulating Brownian motion with drift and volatility
for (i in 2:5573)
  BM[i]<- mu.hat+BM[i-1] + sigma.hat*rnorm(1)
#computing values for geometric Brownian motion
GBM<-price[1]*exp(BM)
#plotting actual and simulated trajectories
plot(Date, price, type="l", lty=1, lwd=2, col="red3",
     xlab="Time", ylab="Netflix Stock price")
lines(Date, GBM, lwd=2, col="lightblue")
legend("topleft", c("Actual price", "Simulated price"),
       lty=1, col=c("red3", "lightblue"))
```

b) Results


Netflix Stock price
0
200
400
600
2005
2010
2015
2020
2025
Year


Frequency
0
1000
2000
3000
-0.2
-0.1
0.0
0.1
0.2
Log-price increments

```
> print(mu.hat<- mean(log.ratio))
[1] 0.001132
> print(sigma.hat<- sd(log.ratio))
[1] 0.03554163
```



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Krupal Patel (2002-2024). (2024, August). *NetFlix Stock Data*. Kaggle. Retrieved December 2nd, 2024.. https://www.kaggle.com/datasets/krupalpatel07/netflix-stock-data

Yahoo Finance (2024) Netflix, Inc calls table, Retrieved December 5th 2024 9:23 pm. https://finance.yahoo.com/quote/NFLX/options/?strike=950